Proposal for a new data model

Nubank

Introduction

Problem Overview

We need to redesign the current data model to better support Nubank's growth.

- Nubank is expanding into new countries
- New products are constantly being launched (e.g., lending, insurance, rewards)
- Not all products are related to peer-to-peer transactions
- Some products are country-specific

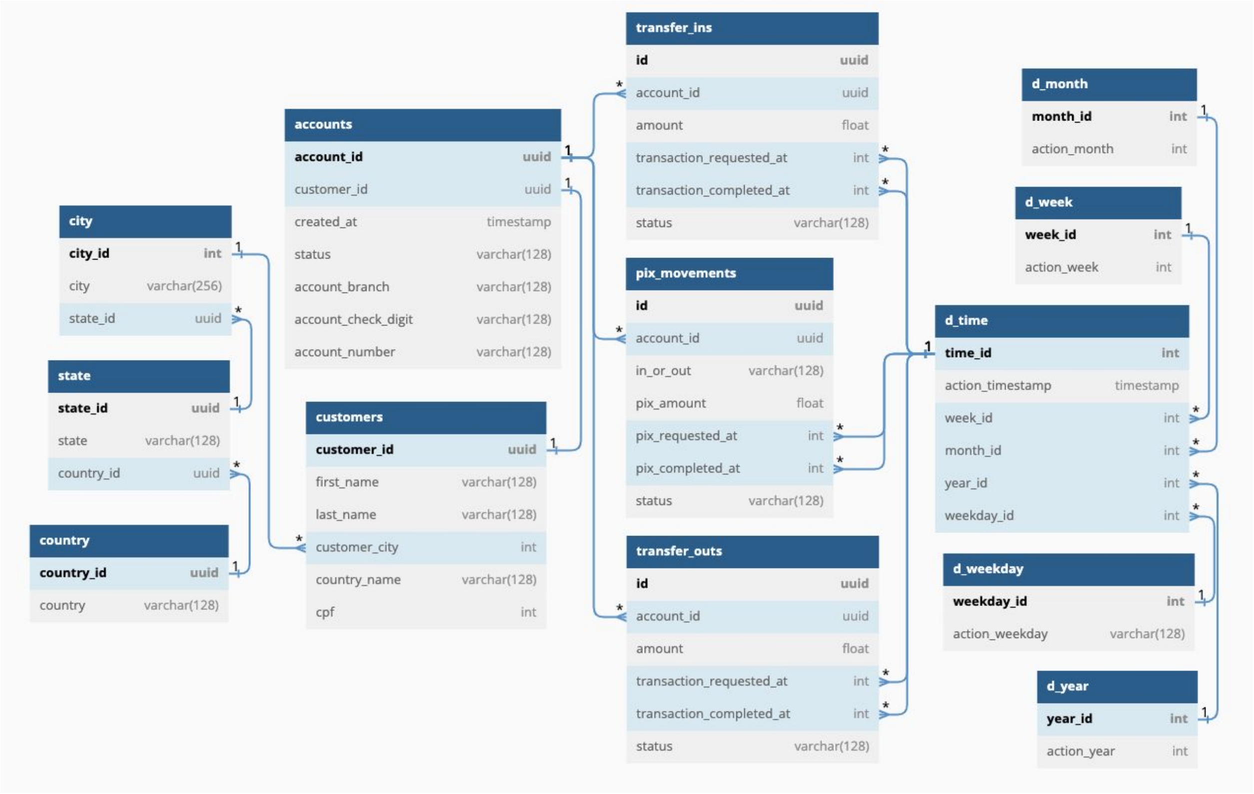
The current model must evolve to handle this increasing complexity.

Objective

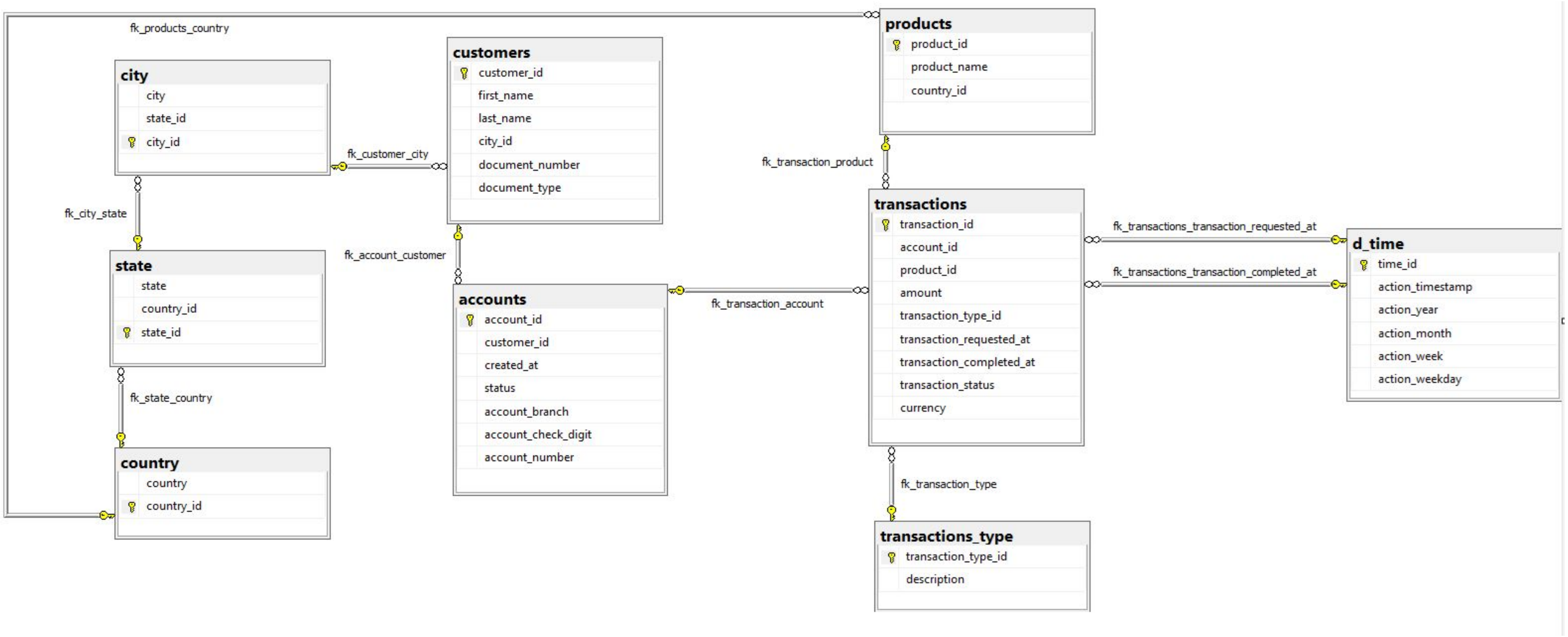
Propose an improved data model that:

- Supports multiple products and countries
- Is scalable for future growth
- Is consistent, clear and easy to use for analysts
- Balances technical quality and business usability

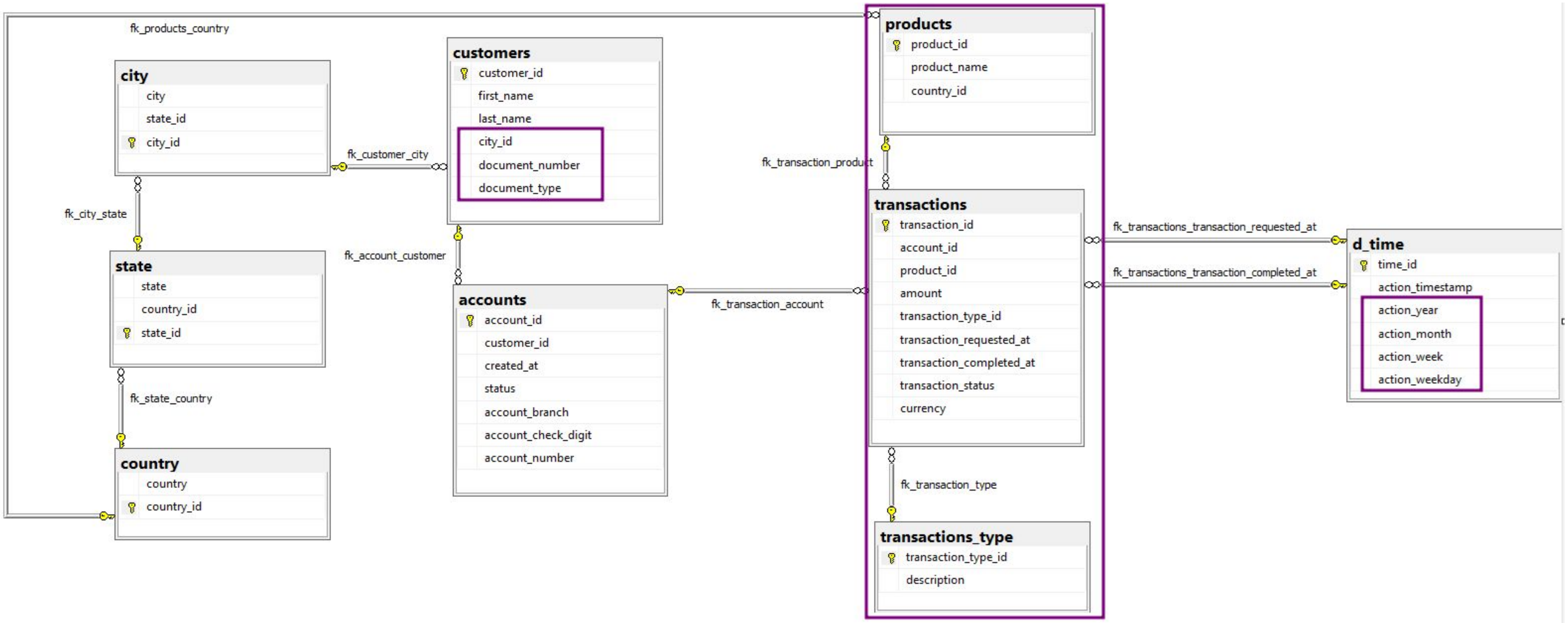
Current Data Model



Proposed Data Model



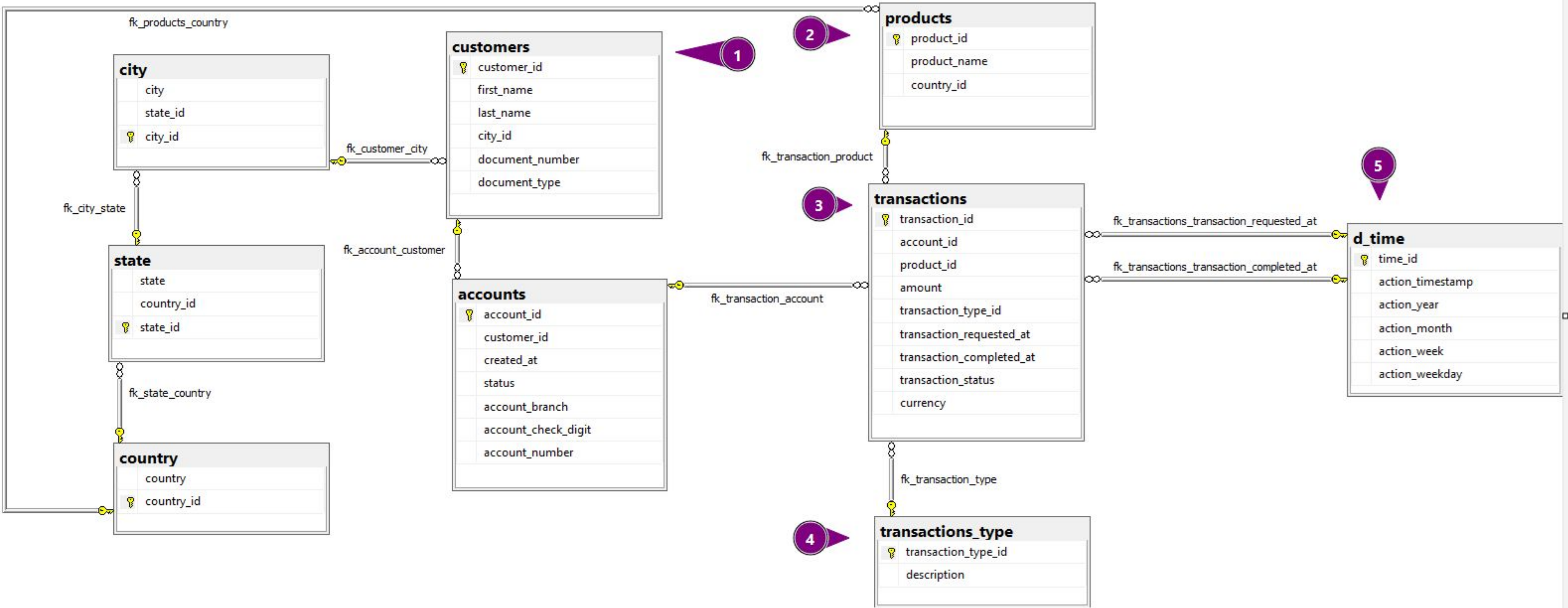
Proposed Data Model (highlighting main changes)



What changes?

What changes?

Considering the numerical main changes as follows



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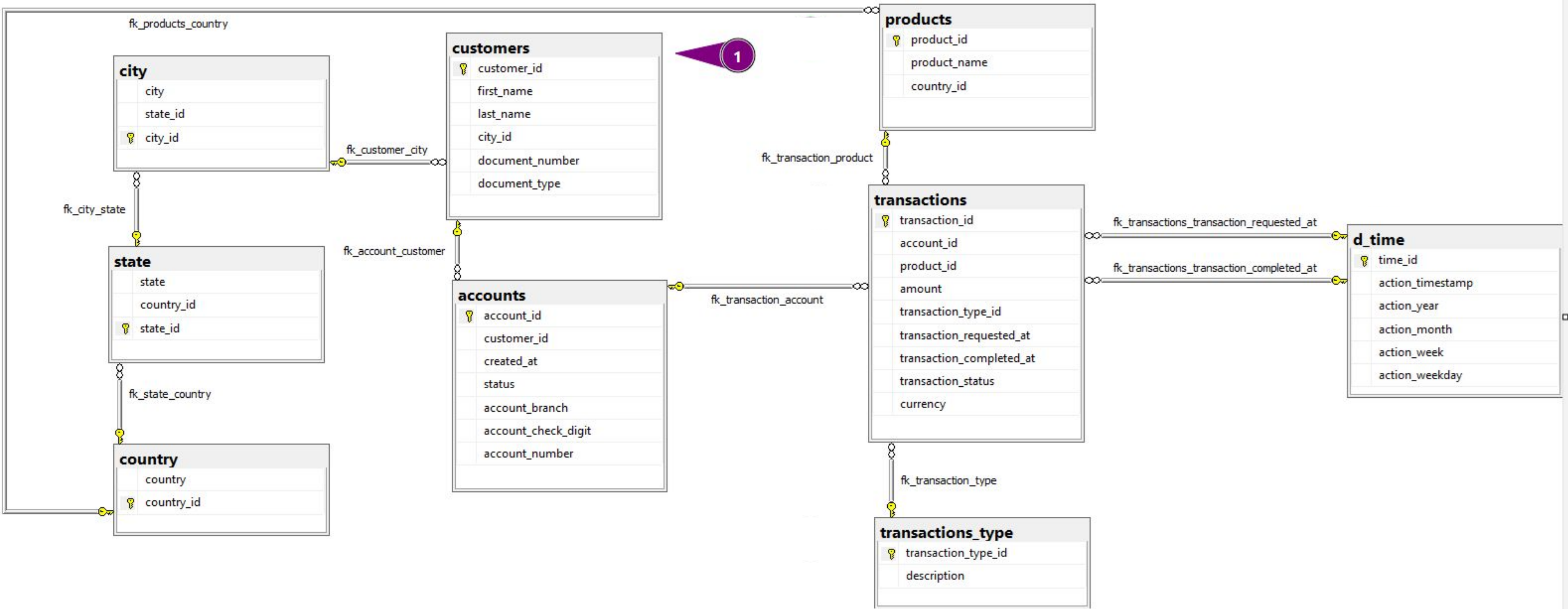


Customers table

This is an existing table

What was removed?

- The 'country_name' column has been deleted
  - ***Objective*** 1) Eliminate redundancy, as country information can be query from the relationships between city, state, and country tables; 2) Avoid future inconsistencies caused by adding new countries and/or non-standardized country values (e.g., multiple representations of the same country).

What was modified?

- The name of the field ‘customer_city’ has been changed to ‘city_id’
  - ***Objective*** 1) Clarify that the field is a foreign key referencing the ‘city’ table; 2) Ensure consistency with naming conventions.
- The name of the field ‘CPF’ has been changed to ‘document_number’
  - ***Objective*** Support internationalization by allowing storage of different document numbers (e.g., Social Security Number, Passport, etc.).

What was added?

- New field ‘document_type’, to reference whether it is a CPF (Brazilian Individual Taxpayer Registry) or another type of document.
  - ***Objective*** 1) Enable identification of the document type (e.g., CPF, Passport, etc.); 2) Support international expansion with multiple document types.

proposed data model

Customers table

Points to consider

Before making any changes, it is **critical** to validate that:

- The ‘CPF’ field is not used in validation logic (e.g., checksum or format validation).
- The ‘CPF’ and/or ‘country_name’ fields are not required by upstream or downstream systems.

Risk:

- Changes to these fields without proper validation could break integrations or negatively impact existing business processes.

proposed data model

Customers table

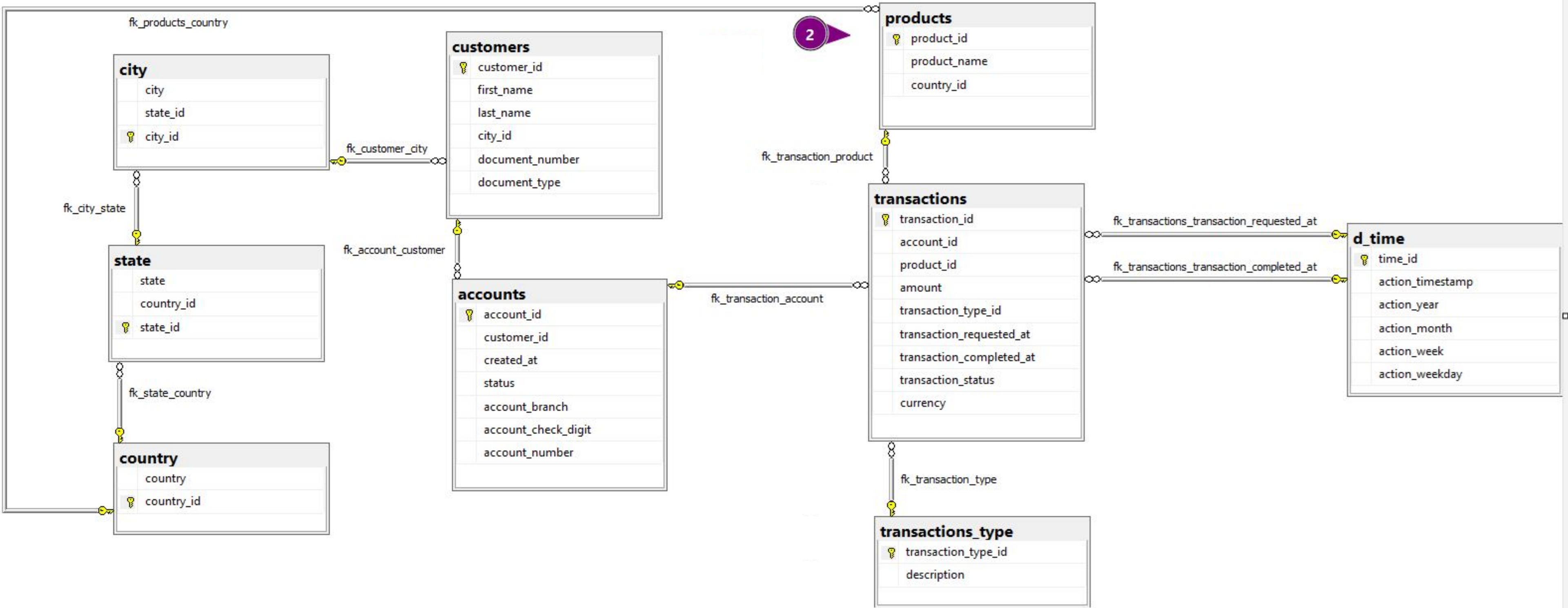
Sample data after modifications

customer_id	first_name	last_name	city_id	document_number	document_type
158554461894233	Rosemary	Haynes	336249655379697984	61733221860	CPF
1195719066794785	David	Gibson	3065955382848678912	3061598443	CPF
1224030529509161	Marlene	Greenfield	1298948914652025344	95690672593	CPF
1643026787581571	Antoine	Free	1580868134171330304	261149495	CPF
2729188736308849	Stacy	Laymon	648580340723423232	89698937168	CPF
...	...	...	...	...	...
810736911063121920	Phyllis	Nelson	903515599400755072	23395725137	CPF
810906841569322240	Toni	Markham	1962889254359325952	62289346069	CPF
811136329041355776	Kiley	Orphey	1298948914652025344	80822539852	CPF
811508290345381632	Pia	Smith	2365744306120669184	62687072305	CPF
811617788423618816	Tasha	Allen	1671822576895199744	24830745092	CPF

proposed data model

What changes?

Considering the numerical main changes as follows



Products table

This is a new table

Why this table was added?

Combined with the **transactions** table, it eliminates the need to create separate tables for each transaction/product type (e.g., pix movements, transfer ins, transfer outs). This design supports the addition of new products over time

What was added?

- New field ‘product_id’, primary key to uniquely identify the products
- New field ‘product_name’, to storage the name of the product (e.g., Pix Movements, Transfer Movements)
- New field ‘country_id’, foreign key referencing to the ‘country’ table, to link a product to it’s country.
    - ***Objective*** As Nubank plans to expand to other regions, this field allows for greater flexibility in the link between country and product.

proposed data model

Products table

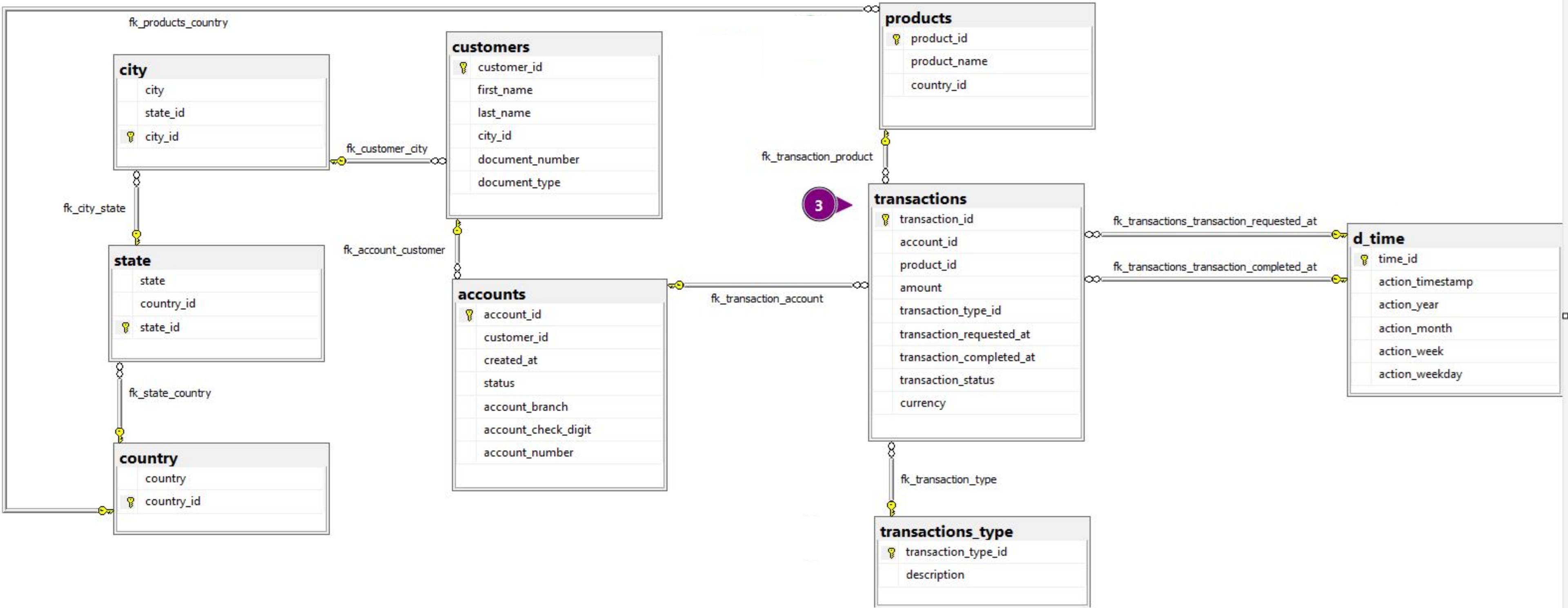
Sample data after modifications

product_id	product_name	country_id
1	Pix Movements	1811589392032273152
2	Transfer Movements	1811589392032273152

proposed data model

What changes?

Considering the numerical main changes as follows



Transactions table

This is a new table

Why this table was added?

This table consolidates the previous tables ‘pix_movements’, ‘transfer_ins’, and ‘transfer_outs’, which have been removed of the data model.

Combined with the **products** and **transaction type** tables, it eliminates the need to create separate tables for each transaction/product type.

What was modified?

- The name of the field ‘id’ has been changed to ‘transaction_id’
  - ***Objective*** 1) Clarify that the field is the primary key of the table; 2) Ensure consistency with naming conventions.
- The name of the field ‘status’ has been changed to ‘transaction_status’
  - ***Objective*** Ensure consistency with naming conventions.

What was added?

- New field ‘product_id’, foreign key referencing to the ‘products’ table, to link a transaction with a product (e.g., Pix Movements, Transfer Movements)
- New field ‘transaction_type_id’, foreign key referencing to the ‘transaction_type’ table, used to classify the transaction type (e.g., Pix In, Pix Out, Transfer In, Transfer Out)
- New field ‘currency’, for as Nubank expands to other countries and products, it’s important to know the currency associated with each transaction. This ensures accurate financial reporting.

proposed data model

Transactions table

Points to consider

- Check if Nubank really intends to start operating in multiple currencies, because if so, it's worth having a separate currencies table in the future.

Transactions table

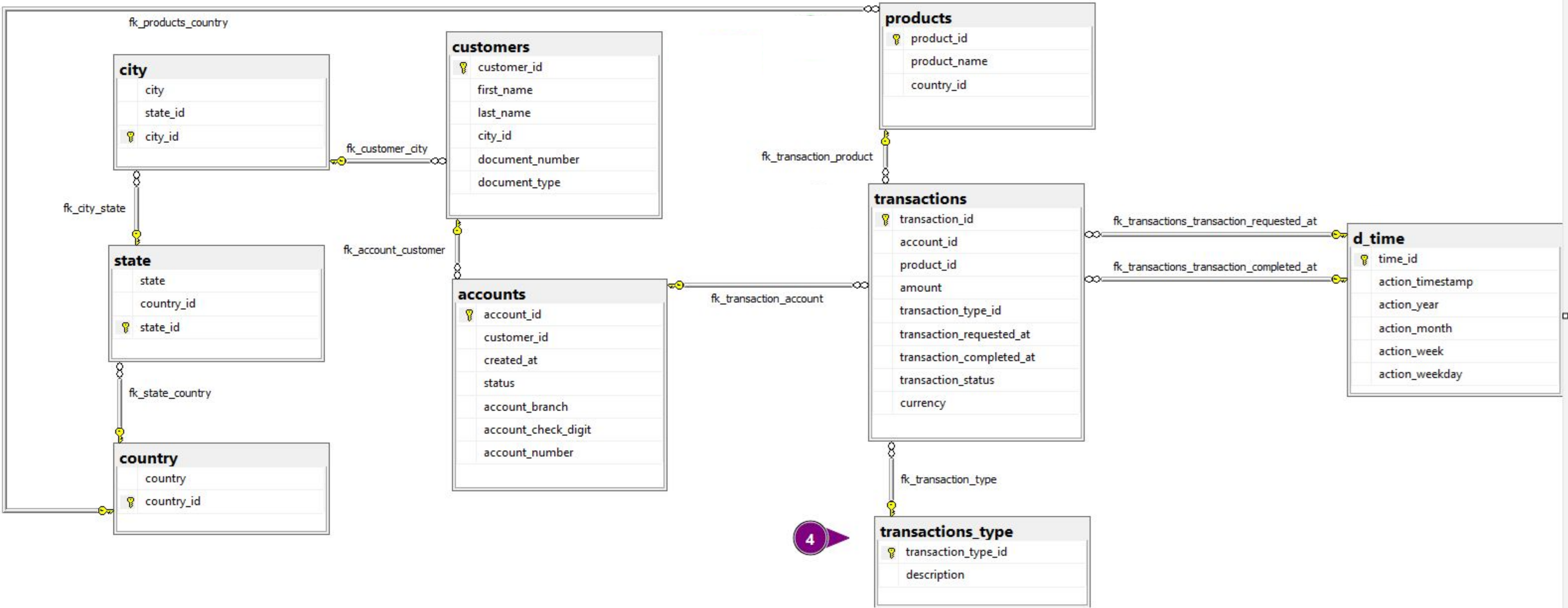
Sample data after modifications

transaction_id	account_id	product_id	transaction_type_id	transaction_requested_at	transaction_completed_at	transaction_status	amount	currency
1905526228328	3344350342360918528	2	21	1592237266	NaN	failed	1825.14	BRL
8050018111488	436078529800937088	1	11	1589037217	1.589037e+09	completed	1982.46	BRL
15468786971758	3386401529082005504	2	21	1609200019	1.609200e+09	completed	1866.63	BRL
21047734532096	196118294852870752	2	21	1606417136	1.606417e+09	completed	448.48	BRL
21538059756947	2049854582898863360	2	22	1579355316	1.579355e+09	completed	1797.07	BRL
...	...	...	...	...	...	...	...	...
3402811221264279552	49447840776902688	2	22	1597266514	1.597267e+09	completed	444.20	BRL
3402811438383354880	3251023429694187008	1	11	1607396825	1.607397e+09	completed	1390.47	BRL
3402817623509785600	1670685025093598208	2	21	1603383090	1.603383e+09	completed	840.20	BRL
3402819480072835072	3189881699920608768	1	11	1584020496	1.584020e+09	completed	388.77	BRL
3402821787374119424	1445575213760933120	2	21	1597197793	1.597198e+09	completed	416.68	BRL

proposed data model

What changes?

Considering the numerical main changes as follows



Transactions Type table

This is a new table

Why this table was added?

Combined with the **transactions** table, it eliminates the need to create separate tables for each transaction/product type (e.g., pix movements, transfer ins, transfer outs).

What was added?

- New field ‘transaction_type_id’, primary key to uniquely identify the transaction types
- New field ‘description’, to storage the description of the transaction type (e.g., Pix In, Pix Out, Transfer In, Transfer Out).

proposed data model

Transactions Type table

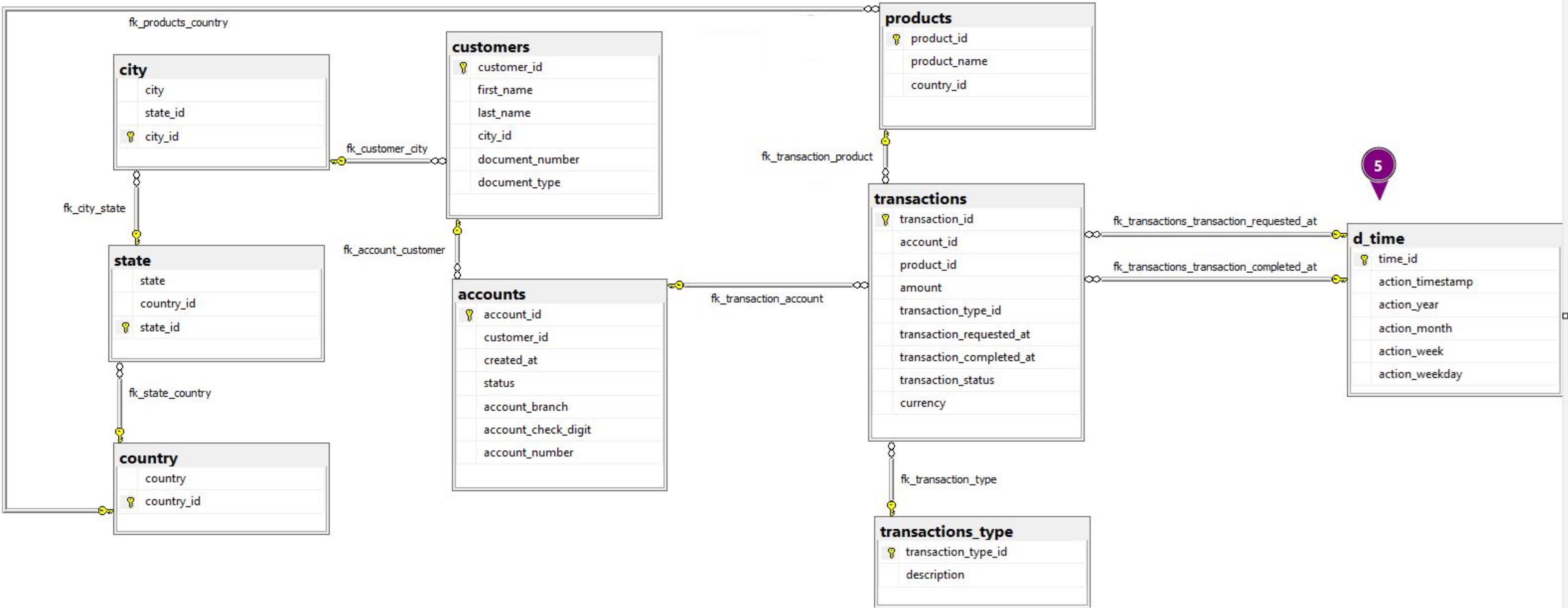
Sample data after modifications

transaction_type_id	description
11	pix_in
12	pix_out
21	transfer_in
22	transfer_out

proposed data model

What changes?

Considering the numerical main changes as follows



Time table

This is an existing table

What was removed?

- The auxiliary tables for d_time were removed: ‘d_week’, ‘d_month’, ‘d_weekday’ and ‘d_year’
- The columns ‘week_id’, ‘weekly_id’, ‘monthly_id’, and ‘year_id’ were also removed.
  - ***Objective*** The auxiliary time tables can be removed from the data model, as d_time already provides the necessary information through **action_timestamp,** which can be queried as a date field using standard date functions, and the action columns can be added as new fields, simplifying the model without losing functionality.

What was added?

- New columns for ‘action_year’, ‘action_month’, ‘action_week’ and ‘action_weekday’, data that were derived from the tables removed, now can be query without multiple joins.

proposed data model

Time table

Sample data after modifications

time_id	action_timestamp	action_year	action_month	action_week	action_weekday
1577836802	2020-01-01T00:00:02.000Z	2020	1	202000	2
1577836805	2020-01-01T00:00:05.000Z	2020	1	202000	2
1577836834	2020-01-01T00:00:34.000Z	2020	1	202000	2
1577836844	2020-01-01T00:00:44.000Z	2020	1	202000	2
1577836896	2020-01-01T00:01:36.000Z	2020	1	202000	2
...	...	...	...	...	...
1609459041	2020-12-31T23:57:21.000Z	2020	12	202052	3
1609459054	2020-12-31T23:57:34.000Z	2020	12	202052	3
1609459062	2020-12-31T23:57:42.000Z	2020	12	202052	3
1609459093	2020-12-31T23:58:13.000Z	2020	12	202052	3
1609459106	2020-12-31T23:58:26.000Z	2020	12	202052	3

proposed data model

What remains the same?

What remains the same?

Tables ‘country’, ‘state’, ‘city’ and ‘accounts’ did not changed

- The country table is already in the ideal format to accommodate new countries as Nubank expands.
- Separating country, state, and city data into separate tables is useful for individual use; for example, in 'customers' we need to know the customer's city, while in 'products' we only need to know the product's country.
- Storing country, state, and city data in specific tables is ideal for avoiding data standardization issues.
- No problems were identified in the table of accounts, neither in it’s construction nor in the way it is used, as described in the use case.

proposed data model

Mapped changes & Attachments

Mapped changes

Guideline for changes in current data processes for the technical teams

Old Table	Old Name Column	New Table	New Name Column
customers	country_name	country	country
customers	customer_city	customers	city_id
customers	cpf	customers	document_number
d_time	week_id	d_time	-
d_time	weekday_id	d_time	-
d_time	month_id	d_time	-
d_time	year_id	d_time	-
d_year	action_year	d_time	action_year
d_month	action_month	d_time	action_month
d_week	action_week	d_time	action_week
d_weekday	action_weekday	d_time	action_weekday

Mapped changes

Guideline for changes in current data processes for the technical teams

Old Table	Old Name Column	New Table	New Name Column
pix_movements	id	transactions	transaction_id
pix_movements	account_id	transactions	account_id
pix_movements	pix_amount	transactions	amount
pix_movements	in_or_out	transactions_type	description
pix_movements	pix_requested_at	transactions	transaction_requested_at
pix_movements	pix_completed_at	transactions	transaction_completed_at
pix_movements	status	transactions	transaction_status
transfer_ins	id	transactions	transaction_id
transfer_ins	account_id	transactions	account_id
transfer_ins	amount	transactions	amount
transfer_ins	transaction_requested_at	transactions	transaction_requested_at
transfer_ins	transaction_completed_at	transactions	transaction_completed_at
transfer_ins	status	transactions	transaction_status

Mapped changes

Guideline for changes in current data processes for the technical teams

Old Table	Old Name Column	New Table	New Name Column
transfer_outs	id	transactions	transaction_id
transfer_outs	account_id	transactions	account_id
transfer_outs	amount	transactions	amount
transfer_outs	transaction_requested_at	transactions	transaction_requested_at
transfer_outs	transaction_completed_at	transactions	transaction_completed_at
transfer_outs	status	transactions	transaction_status

You can refer to the file **mapped_changes_nubank.xlsx** for the full list

Attachment

You can refer to the file **jupyter_notebook_nubank.ipynb** for the full development logic, including:

For the current model

- CSV ingestion into dataframes
- Initial exploratory data analysis
- Export of dataframes to tables in a SQL Server database
- Definition of relationships between tables

For the proposed model

- Setup of a homologation environment
- Documented DDL in the homologation environment
- Modifications to table relationships in the homologation environment

Any questions?

Thank you!